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Kalman Filter Design for Fast Synchronization of a High-Stability Rubidium Oscillator

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***Abstract*—The article briefly describes the methodology of designing a Kalman filter that it is used in the phase lock loop (PLL). At the beginning the tools and the experimental environment are discussed, especially the structure of the synchronization loop. First, the characteristic of the 1PPS GNSS reference signal is obtained using superposition by gathering data when the oscillator is locked to GPS. This data is used to calculate a covariance matrix of the measurement noise for the Kalman filter. Next, a signal without noise from a very stable frequency source is applied as a reference signal, and the identification experiment of the oscillator is executed. Based on the identification data, one-, two-, three- and four-dimensional space-state models of the rubidium clock are created and the process noise covariance matrix for the Kalman filter is calculated. At the end, the quality of Kalman filter tracking is examined. For this purpose, MSE between the real behavior of the phase and the estimation provided by the filter is calculated. Finally, the optimal dimensionality of the model is discussed. It is also shown how much the estimations of the phase prediction in Kalman filter can be trusted and how to adjust the process noise covariance matrix properly.**

Keywords-Kalman filter; synchronization; rubidium oscilator; GPS; synchronization loop

I. INTRODUCTION

Nowadays technical solutions require an increased number of precise time and frequency sources to synchronize the whole system. A well-known technique to obtain a high-quality frequency signal is to synchronize a local oscillator, e.g., a rubidium clock to the pulse per second (1PPS) signal obtained from the global navigation satellite system (GNSS). The majority of researchers focus on the improvement of long-term properties of a synchronization loop by modifying loop filters [1] and control algorithms. In this article, a different approach is shown, since in some applications it is necessary to make fast adjustments of phase or frequency, e.g., when a failure has occurred in the system and the synchronization is lost. In such a case, a different design is necessary to provide fast synchronization [2].

The reduction of measurement noise is a key factor in reducing synchronization time. The simplest solution that significantly lowers the noise level is moving average (MAV) filtration, but increasing the length of the filter simultaneously increases signal delay. For signal tracking in the presence of additive white Gaussian noise, the optimal solution is Kalman filtration, but a faulty model or inappropriate parameters, can result in a steady state-error.

This article introduces a design of the Kalman filter for fast synchronization of a high stability rubidium oscillator. For this purpose a complete laboratory set was created. The synchronization loop was implemented. The elements chosen for experiments, such as the rubidium clock, GPS receivers, FPGA, microcontrollers etc., are commercially available and commonly used in industry. The physical data gathered in the measurements during experiments were processed in MATLAB. Because of the empirical approach, the obtained results have practical importance.

The article is organized as follows: Section II describes the structure of the synchronization loop, in Section III the methodology of GPS measurements is presented, in Section IV models of the oscillator are calculated and Section V gives a general description of Kalman filtration. Section VI contains the experimental results and conclusions are presented in Section VII.

II. SYNCHRONIZATION LOOP DESCRIPTION

The synchronization of a highly stable oscillator is commonly realized by the synchronization loop, whose functional diagram is presented in Fig. 1. The REF element is a reference source, which can be a signal from the oscillator with better properties than the one used in the synchronization loop, or it can be the 1PPS signal from GNSS. The 1PPS signal indicates the beginning of each second triggered by an atomic clock placed on a satellite. The time signal is transmitted to the earth, but during propagation through the atmosphere it suffers from various delays. In theory, when we are averaging 1PPS for a long time, the mean value of its period equals exactly one second. In practice, for various receivers [3] the mean value of the period is obtained exact to a few picoseconds and the standard deviation varies from a few nanoseconds to over a dozen nanoseconds for different devices provided by different manufacturers. Usually a 1PPS signal also has Gaussian distribution.

The phase detector (PD) measures the phase difference between the reference signal and the signal from the local oscillator. It is a time-to-digital converter that converts an analog time shift between two slopes to a digital number. In our experiment the phase detector was implemented in FPGA (Xilinx Spartan

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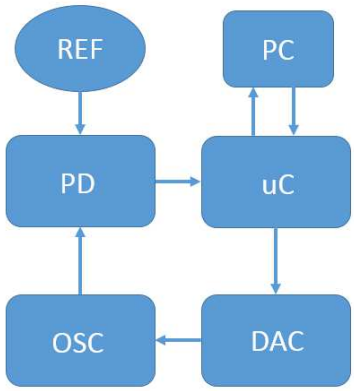

Figure 1. Synchronization loop schematic diagram

6 XC6SLX9) as a binary counter. The rising slope of the reference signal enables the counting procedure of the clock ticks of the internal FPGA clock working with the frequency of 300 MHz. The rising slope of the signal of a local oscillator disables counting and each obtained tick equals 3.33 ns of the phase difference.

Information about the phase is sent to a microcontroller (uC), which in our solution is ATxmega256A3BU. This microcontroller supports USB communication and a virtual serial port. Therefore, information about the phase can be easily sent to MATLAB and then processed in a high-level programing language. The communication via a virtual serial port between a computer (PC) and the microcontroller is bidirectional, so ASCII commands can also be sent from MATLAB to the microcontroller, e.g., to send a code word to the digital-to-analog converter (DAC).

The synchronization circuit contains a precise 20-bit DAC from Texas Instruments (TI DAC1220E). The output value changes from 0 to 5 volts with a very small step, this way a fine adjustment of the synchronization loop is possible.

In the test circuit, an LPFRS rubidium clock was used (denoted as OSC). The output frequency of the oscillator is 10 MHz, and it can be precisely adjusted in the range of $\pm 1\text{x}10^{-9}$ by changing the analog control voltage. For 2.5 V, the oscillator works with the nominal frequency, therefore, for 0 V the frequency is maximally reduced and for 5 V it is maximally increased. A change of the frequency by $1\text{x}10^{-9}$ means that every second the phase of the oscillator shifts by 1ns with reference to the original frequency.

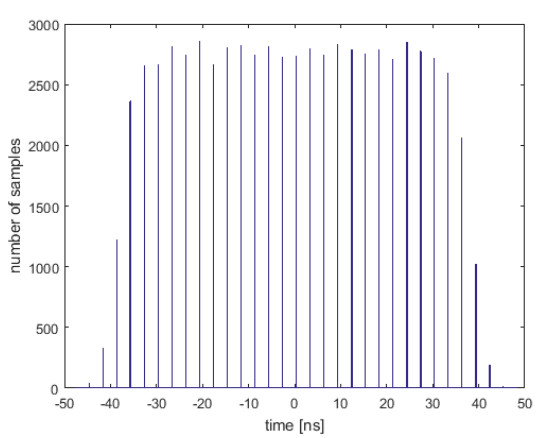

Figure 2. Trimble Acutime GG PPS empirical histogram of the phase difference beetween the local oscilator recorded on the phase detector

III. GPS Measurement

In order to characterize the signal from GPS and to estimate the measurement noise, an experiment was carried out. The system presented in Fig. 1 has been used for this purpose. In the described application, a 1PPS signal from the GPS receiver was applied as the reference source. Then a normal synchronization process was started. After some time the synchronization loop was in a steady state, which means that the phase difference was near zero and the code word was almost constant. In other words, the phase and frequency of the local oscillator were synchronized. Furthermore, to make a fine adjustment of the oscillator, a PI regulator with low gain was used as a control algorithm. After the local oscillator was locked to the reference GPS signal, only very small adjustments of the clock frequency were made. As the rubidium clock is very stable and has good long-term properties, like low noise, in the synchronization state all measurements coming from the phase detector were mostly affected by the GPS noise, the path noise and the conversion noise. In Fig. 2, a histogram of the measured phase error in the steady state is presented.

In the described case Trimble Acutime GG [4] was used as the GPS receiver. According to the information provided by the manufacturer it has 1 sigma accuracy equal to ±15 ns. For the data presented in Fig. 2, the standard deviation was equal to 21.47 ns. Assuming 15 ns from the receiver and 3.33 ns from the converter there were about 3 ns of unpredicted disturbances. The histogram bars are spaced by 3.33 ns because of the converter resolution. In the conducted experiments a more precise GPS receiver provided by U-blox LEA-6T [5] has also been used. In the second case, the measured standard deviation was equal to 12.43 ns.

IV. Rubidium Oscillator Space State Model

To describe the behavior of a real system, mathematical abstraction models were created. A commonly used model is a space state model given by equations (1,2):

$$x(n+1) = \boldsymbol{A}x(n) + \boldsymbol{B}u(n) + \boldsymbol{K}d(n), \tag{1}$$

$$y(n) = \boldsymbol{C}x(n) + \boldsymbol{D}u(n), \tag{2}$$

where $x(n)$ is a state vector with values reflecting the actual state of a modeled object in the n-step, $u(n)$ is an input vector that contains input values, $y(n)$ is an output vector that contains output values, and $d(n)$ is a disturbance vector. **A** is a square state matrix with as many rows as states and it describes how the state vector changes in one step. **B** is an input matrix with as many rows as states and as many columns as inputs and it provides information how the input vector affects the inner state of the model. **C** is an output matrix with as many rows as outputs and as many columns as states and it translates the inner state to real, physical values. **D** is a feedthrough matrix with as many rows as outputs and as many columns as inputs

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and it describes how a control word directly affects the outputs. **K** is a state disturbance matrix.

The rubidium oscillator is a voltage controlled oscillator (VCO) and therefore the model in the space state has only one input, which is voltage that varies in the range 0-5 V and changes the frequency of the oscillator, and only one output, which is modeled as a phase difference between the reference and modeled oscillators. The question is how many hidden states should the state vector have to properly reflect the dynamics of the object. Therefore, the state vector can be represented as one-state, two-state, three-state or N-state vector (3-6):

$$x(n) = (x_{ph}(n)), \tag{3}$$

$$x(n) = (x_{ph}(n), x_{dph}(n)), \tag{4}$$

$$x(n) = (x_{ph}(n), x_{dph}(n), x_{ddph}(n)), \tag{5}$$

$$x(n) = (x_{ph}(n), xd_{ph}(n), \dots, x_{ndph}(n)), \tag{6}$$

where x_{ph} is mapping the phase of the object, x_{dph} is derivative of x_{ph} and describes how fast the phase is changing – this way the frequency of the object is mapped. x_{ddpd} is derivative of x_{dph} and describes how fast the frequency is changing, etc..

In order to find the numerical values of the matrices in the space state models, an identification experiment for the control object was executed. Therefore, a signal from a highly stable generator with very low noise was applied to the circuit presented in Fig.1 as the reference signal. Then the rubidium clock (LPFRS [6]) was calibrated in such a way that the control voltage at the level of 2.5 V provided frequency synchronization with the reference, which means that the phase error remained constant. Then, every 30 s the voltage value was changed from 1 V to 4 V and back from 4 V to 1 V. This rectangular control signal resulted with a triangular phase signal on the output of the phase detector in comparison to the reference signal. Such behavior is consistent with theory, because the clock is an integrating object with inertia. The input and output signals were used to estimate the space state models. By putting numerical values from the experiment as input arguments to the MATLAB function *ssest*() [7], the first, second, third and fourth order models were created.

V. Kalman Filter Principles

Kalman filtration is well known and widely used in many practical applications [8]. This solution is optimal when measurements are affected by an additive white Gaussian noise. The filtration algorithm consists of two successively repeated phases. In the prediction phase, the predictions of the new state vector and process matrix are made. The prediction can be expressed by the following equations (7-8):

$$x_{pred} = \boldsymbol{A}x(n) + \boldsymbol{B}u(n), \tag{7}$$

$$\boldsymbol{P}_{pred} = \boldsymbol{APA}^T + \boldsymbol{Q}, \tag{8}$$

where **P** is a process matrix and $\mathbf{P}_{pred}$ is a prediction of this matrix, x_{pred} is a prediction of the state vector and **Q** is a process noise covariance matrix. **P** and **Q** are square matrices with as many rows as states. Other symbols are the same as in (1,2) because the Kalman filter is based on the space state model. When the prediction phase is over, the Kalman gain **K** is calculated (9):

$$\boldsymbol{K} = \boldsymbol{P}_{pred}\boldsymbol{C}^T\left(\boldsymbol{CP}_{pred}\boldsymbol{C}^T + \boldsymbol{R}\right)^{-1}, \tag{9}$$

where **R** is a measurement noise covariance matrix with as many rows and columns as inputs. Then, using **K,** proper corrections are calculated (10-11):

$$x(n+1) = x_{pred} + \boldsymbol{K}\left(z - \boldsymbol{C}x_{pred}\right), \tag{10}$$

$$\boldsymbol{P} = (\boldsymbol{I} - \boldsymbol{KC})\boldsymbol{P}_{pred}, \tag{11}$$

where z is a vector of the measured values. When **K** values are low, the algorithm trusts more in prediction and when **K** values are high, the output is mostly based on measurements. **P** transfers information about the process between iterations.

The design of the Kalman filter is about finding all matrices. The **A**, **B** and **C** can be obtained by an identification experiment. Process matrix **P** has a property of auto-adjustment, therefore it can be set initially as an identity matrix. Measurement noise covariance matrix **R** can be determined from the input data by calculating the standard deviations of the noise for each input. Usually the problem lies in finding the optimal value of **Q**. This issue has been addressed in the following section.

VI. Experimental Results

The comparison experiment was organized as follows. First, the synchronization procedure to a high-quality reference source was conducted. The initial phase error in the loop was equal to 50 ns and for 600 s, a synchronization circuit was observed. The values of the phase error trajectory and control words on DAC were saved and used as the reference phase trajectory. Noise from the GPS measurement (Section III) was added to the designated trajectory in MATLAB. The Kalman filter was applied afterwards and the mean squared error (MSE) between the reference trajectory and filtered trajectory was calculated.

For fast synchronization, the loop filter should provide maximal noise reduction and cause minimal delay. To compare one-, two-, three- and four-dimensional models, obtained from the identification experiment used to design the Kalman filter, the first process noise covariance matrix **Q** for each model needs to be created. The initial value is calculated from the space state disturbance matrix **K** (12):

$$\boldsymbol{Q} = \boldsymbol{KK}^T. \tag{12}$$

Because for each model the **Q** matrix has a different size, scaling factor e is introduced to eq. 8 (13) to simplify the filter adjustment procedure:

$$\boldsymbol{P}_{pred} = \boldsymbol{APA}^T + e\boldsymbol{Q}. \tag{13}$$

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Fig. 3 shows the behavior of the Kalman filter with different numbers of dimensions in the presence of noise from a U-blox GPS receiver. When the *e* parameter is high, the Kalman filter output is mostly based on measurements and the step response is fast. When the *e* parameter is low, the output of the filter is generally based on predictions and the input noise is strongly reduced, but the step response is slow. Fig. 4 indicates that for every model it is possible to find *e* that provides minimal MSE. When the *e* value is near zero, the output of the filter is based purely on predictions and in this case it is clearly visible that the three-dimensional model yields the best performance. The one-dimensional model is undermodeled and the four-dimensional one is overmodeled. When the *e* value provides minimal MSE it achieves the best tradeoff between the speed and noise reduction. Only for the three-dimensional model lowering the *e* value from the optimal point in order to achieve further noise reduction is still possible without losing the tracking ability.

In Fig. 4 the behavior of different dimensional Kalman filters in the presence of noise from a Trimble Acutime GG GPS receiver is shown. The space state models are exactly the same as in Fig. 3, only the input noise has changed. Because the input noise is high, the one-dimensional model provides the lowest MSE. This situation occurs, e.g., in IoT devices, when a low quality GPS receiver is applied and the usage of the one-dimensional model is favorable, because 1x1 matrices are multiplied, not 2x2 or 3x3 ones, so the reduction of computational complexity is obtained and the overall power consumption of the device is decreased.

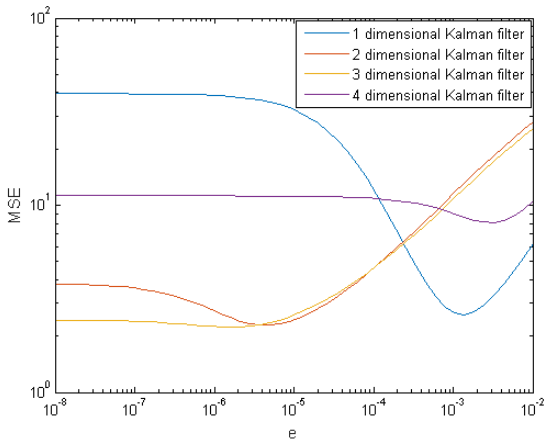

Figure 3. Comparison of MSE characteristics as a function of the scaling factor for low measurement noise (U-blox LEA-6T)

VII. Conclusions

In this article presents the complete methodology of designing the Kalman filter for fast synchronization. One-, two-, three- and four-dimensional models of the rubidium clock were examined. There is no simple indicator of the model which yields the best performance. The one-dimensional model has low computational complexity, which is important for small and cheap microcontrollers. When the consumption of resources is not important, it is better to use the two- or three-dimensional model. As shown by the results, higher dimensionality does not provide any further improvement of the filter performance. It is also worth noting that when the measurement noise is high modeling impact on the performance of the filter is lower.

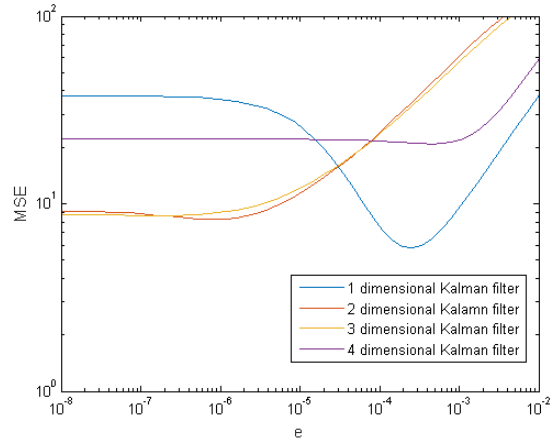

Figure 4. Comparison of MSE characteristics as a function of the scaling factor for high measurement noise (Trimble Acutime GG)

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